

■ AlertNex — Smart India Hackathon 2026

Official Team Presentation Master Script & Comprehensive Viva Q&A; Dossier

Problem Statement: **SIH26001** | Ministry: **Ministry of Development of North Eastern Region (MDoNER)**

Team: **AlertNex** | Leader: **Ayush Kumar**

1. TEAM ROSTER, ROLES & TECHNICAL OWNERSHIP BREAKDOWN

Har member ko exact pata hona chahiye ki unhone project me **kya banaya hai**, kaun si **technologies use ki hain**, aur presentation me **kya bolna hai**. Is table se poora team collaboration transparent aur authentic lagega:

Member & Official Role	Technical Modules Built & Technologies Used	Stage Presentation Role
1. Ayush Kumar <i>Team Leader</i>	- System Architecture & End-to-End Orchestration - Multi-source Risk Fusion Formula & Pipeline Tech: Python, System Design, XAI Logic	Opening hook, problem statement crisis, project vision, architecture summary & closing pitch.
2. Prerana Mondal <i>Frontend & UI/UX Lead</i>	- Design System (Forest Green, Topo contours, Light Cards) - Real-time Telemetry Dashboard & KPI Grid Tech: HTML5, CSS3 Tokens, Chart.js, Mobile Drawer	Dashboard walkthrough, UI/UX philosophy, real-time command metrics & mobile responsiveness.
3. Sondeep Kumar <i>Backend & DB Lead</i>	- FastAPI Async REST Engine & Schema Models - Topological Road Graph Solver & Bypass Route Tech: FastAPI, Python 3.12, SQLite/PostGIS, Graph Theory	Backend API throughput, topological connectivity solver, Dijkstra road bypass engine & DB schema.
4. Shinjini Lohar <i>AI/ML & XAI Lead</i>	- Dynamic 5-Factor Landslide Risk Assessment - Explainable AI (XAI) Factor Attribution Engine Tech: Scikit-Learn, NumPy, XAI Weights, Feature Scaling	ML risk calculation formula, feature weights, explainability (XAI) attribution & false-alarm control.
5. Subham Kumar Modi <i>GIS & Field Sync Lead</i>	- Interactive GIS Map Layering & Risk Zone Polygons - Offline-First IndexedDB Ground Incident Reporting Tech: Leaflet.js, OpenStreetMap, IndexedDB API	GIS interactive map, Meghalaya/Sikkim zone risk simulation & zero-connectivity offline sync demo.
6. Rahul Deo <i>DevOps & Security Lead</i>	- Cloud Edge CDN Deployment, CI/CD Pipeline - Production Security, CORS Whitelisting, Cache Control Tech: Netlify, GitHub Actions, Docker Compose, Git	Live cloud infrastructure, edge caching, disaster-grade failover, CI/CD pipeline & data encryption.

2. EXACT PRESENTATION SPEECH SCRIPTS (HAR MEMBER KYA BOLEGA)

Presentation ke dauran har member ko confident, crisp aur technically sound bolna hai. Yahan Hindi + English mix script di gayi hai jo SIH jury ko sabse jyada impress karti hai:

1. Ayush Kumar (Team Leader) — Pitch & Opening (0:00 - 1:30)

"Respected Judges & Jury Members, a very warm good morning. I am Ayush Kumar, Team Leader of AlertNex.

North Eastern Region of India me landslides har saal hazaron lives aur economic lifelines ko disrupt karte hain. NH-10 Sikkim aur NH-29 Nagaland jaise critical national highways par cloudburst ke baad sudden debris flow hota hai, jisse remote villages completely cut-off ho jaate hain. Current government systems mostly **reactive** hain — yaani disaster hone ke baad relief pohanchti hai.

Hum introduce kar rahe hain **AlertNex**: Ministry of DoNER ke liye India ka pehla **AI-Powered Early Warning, Explainable Risk Assessment aur Topological Connectivity Decision Support System**. AlertNex na sirf 6 se 12 ghante pehle multi-parameter hazard predict karta hai, balki commanders ko ye bhi batata hai ki kaun sa road band hoga aur kaun sa emergency bypass route open hai. Ab meri team aapko live system ka walkthrough degi."

2. Prerana Mondal (Frontend & UI/UX Lead) — Command Dashboard (1:30 - 2:30)

"Thank you Ayush. Respected Judges, I am Prerana Mondal, Frontend & UI/UX Lead.

Humne AlertNex ka interface ek **Command-Grade Government Dashboard** ke roop me design kiya hai. Iska theme environmental aur terrain identity ko reflect karta hai — Deep Forest Green, Earth Brown aur Crisp Light Content Cards with subtle topographic elevation contours.

Topbar par aap real-time live system status, MDoNER affiliation aur DEMO simulation toggle dekh sakte hain. Main dashboard par instant KPI cards hain: Active Monitored Zones, High-Risk Sectors, Road Disruptions aur Vulnerable Villages. Ye poori UI fully responsive hai — desktop ke saath-saath mobile tablets par frontline disaster officers ke liye optimized hai."

3. Subham Kumar Modi (GIS & Offline Sync Lead) — GIS Map & Field Reporting (2:30 - 3:30)

"Thank you Prerana. Judges, I am Subham Kumar Modi, Geospatial GIS & Field Systems Lead.

Hamara GIS Module Leaflet engine par built hai jo real-time weather stations, rainfall telemetry, soil moisture probes aur historical landslide inventory ko 30-meter elevation model par map karta hai. Yahan aap 8 NER states ke high-risk sectors dekh sakte hain jaise Cherrapunji-Shella, Haflong Pass aur Kohima.

Sabse unique feature hamara **Offline-First Citizen & Field Incident Reporting** module hai. Storms ke dauran jab cellular network blackout ho jaata hai, frontline officers bina internet ke photo aur GPS location ke saath incident log kar sakte hain. Data IndexedDB local storage me store hota hai aur internet restore hote hi background me server se sync ho jaata hai."

4. Shinjini Lohar (AI/ML & XAI Lead) — Multi-Factor AI & XAI Engine (3:30 - 4:30)

"Thank you Subham. Respected Jury, I am Shinjini Lohar, AI/ML & Explainable AI Lead.

Landslides prediction me sabse badi problem 'Black-Box AI' hoti hai — agar model alert deta hai toh DM ko pata nahi chalta ki alert kyu aaya. AlertNex is crisis ko solve karta hai hamare **Explainable AI (XAI) Attribution Engine** ke through.

Hamara model 5 continuous features ko ingest karta hai: 24h Cumulative Rainfall (30% weight), Soil Moisture Saturation (25%), DEM Slope Angle (20%), Historical Slide Frequency (15%), aur Ground Field Reports (10%). XAI engine har hazard class ka exact factor breakdown deta hai — jaise '88% High Risk driven by 145mm rainfall + 82% soil moisture'. Isse administration confident decision leti hai bina false alarm ke darr se."

5. Sondeep Kumar (Backend & Database Lead) — Graph Solver & APIs (4:30 - 5:30)

"Thank you Shinjini. Judges, I am Sondeep Kumar, Backend & Database Architect.

AlertNex ka backend Python FastAPI async framework par built hai jo sub-millisecond latency ensure karta hai. Hamara key mathematical innovation hai **Topological Road Connectivity Graph Solver**.

Hum NER ke critical highway road networks ko mathematical graph (nodes = junctions/villages, edges = road segments with risk weights) ke roop me model karte hain. Jab koi road landslide se block hoti hai, hamara Dijkstra-based algorithm instantly alternate emergency bypass routes calculate karta hai, hospital accessibility time nikaalta hai aur SDRF rescue convoys ke liye safest corridor generate karta hai."

6. Rahul Deo (DevOps, Cloud & Security Lead) — Cloud & Failover (5:30 - 6:15)

"Thank you Sondeep. Respected Jury, I am Rahul Deo, DevOps & Cloud Security Lead.

Disaster management platforms ke liye 99.99% uptime zaroori hai. AlertNex ko Netlify Edge CDN aur Render Cloud par multi-region redundant architecture ke saath deploy kiya gaya hai. Hum automated GitHub Actions CI/CD pipeline use karte hain jo har code commit par unit tests run karke production deploy karti hai.

Security ke liye humne strict CORS whitelisting, immutable client-side caching strategies aur automated alert dispatch channels integrate kiye hain jo high-load monsoons me bhi zero downtime deliver karte hain."

7. Ayush Kumar (Team Leader) — Closing Summary & MDoNER Adoption (6:15 - 7:00)

"To conclude, AlertNex is not just a hackathon prototype — it is an end-to-end, deployable disaster intelligence ecosystem built specifically for the fragile geography of Northeast India.

*Ye MDoNER, State Disaster Management Authorities (SDMA), Border Roads Organisation (BRO) aur local communities ko ek single platform par connect karta hai. AlertNex ke saath hum reactive rescue se **proactive life-saving prevention** ki taraf transition kar rahe hain.*

Thank you very much. We are now open for your questions and live demonstration."

3. TOP 25 TOUGHEST SIH JURY QUESTIONS & WINNING ANSWERS

Ye wo 25 sabse critical sawal hain jo SIH technical judges aur ministry evaluators poochte hain. Har answer ko authentic, structured aur confidence ke saath explain karein:

Q1. Existing systems jaise IMD bulletins ya NDMA portals se AlertNex kaise alag hai?

Answer: IMD bulletins broad regional advisories dete hain (jaise 'Heavy rain in Meghalaya'), jisse local road level par actionable clarity nahi milti. AlertNex 3 key gaps solve karta hai: (1) **Hyper-local 30m resolution risk modeling** combining terrain slope + soil moisture; (2) **Explainable AI (XAI)** jo DM ko alert ka exact mathematical reason batata hai; (3) **Topological Connectivity Solver** jo batata hai kaun sa gaon cut-off hua aur alternative bypass route kya hai.

Q2. Cloudburst ke dauran jab cellular towers down hote hain, tab ground reports kaise aayengi?

Answer: AlertNex me **Offline-First IndexedDB Local Storage Architecture** implement kiya gaya hai. Jab frontline field officers dead network zone me hote hain, app background me camera photo, GPS lat-long aur hazard report device memory me store karti hai. Jaise hi device mobile network ya satellite mesh range me aata hai, Service Worker automatically server ke saath reconcile aur sync kar deta hai.

Q3. Aapka AI Model kaun sa use ho raha hai aur weights kaise calculate hote hain?

Answer: Hamara core AI engine multi-parametric weighted vulnerability matrix use karta hai calibrated against historical geological patterns of NER. Features include: (1) 24h Cumulative Precipitation (30%), (2) Soil Moisture Saturation (25%), (3) DEM Slope Gradient (20%), (4) Historical Landslide Frequency Index (15%), aur (5) Verified Ground Incident Reports (10%). Multi-class thresholds: 0-30% Low (Green), 31-60% Moderate (Yellow), 61-80% High (Orange), 81-100% Critical (Red).

Q4. Explainable AI (XAI) ka real-world disaster management me kya fayda hai?

Answer: Traditional black-box neural networks alert dete hain lekin decision justification nahi dete, jisse District Magistrates multimillion-rupee evacuation order issue karne me hesitate karte hain. AlertNex ka XAI engine har prediction ka dynamic factor attribution provide karta hai (e.g. '78% of risk is caused by 160mm continuous rainfall on a 42-degree saturated slope'). Isse administration confident, auditable aur fast decision le sakti hai.

Q5. Topological Road Connectivity Graph Solver kaise kaam karta hai?

Answer: Hum Northeast ke road network ko mathematical Graph $G = (V, E)$ ke roop me model karte hain jahan Vertices V villages, road junctions aur hospitals hain, aur Edges E connecting highway segments hain. Har edge ka weight travel distance aur dynamic landslide risk score ka function hota hai. Agar koi segment block hota hai, Dijkstra / A* algorithm instantly lowest-risk emergency detour corridor compute karta hai.

Q6. Citizen reports me fake ya spam hazard reports ko kaise filter karte hain?

Answer: Hum 3-tier validation protocol use karte hain: (1) **Spatial Clustering:** Ek hi geographic radius (e.g. 500m) me multiple independent reports ko cluster karke confidence score badhate hain; (2) **Telemetry Cross-Verification:** Agar citizen report aayi hai par rainfall aur moisture normal hai, toh priority flag karke manual verification me daalte hain; (3) **Role-Based Authentication:** Verified SDRF / BRO officers ki reports ko direct high-trust weightage milta hai.

Q7. Backend architecture aur database design kya hai?

Answer: Backend Python 3.12 FastAPI async framework par based hai jo high-concurrency requests handle karta hai. Database layer SQLAlchemy 2.0 ORM ke through PostGIS-enabled PostgreSQL (or embedded SQLite in local prototype) use karta hai jo spatial geospatial queries (ST_DWithin, ST_Contains) ko fast execute karta hai. REST APIs OpenAPI 3.0 compliant hain.

Q8. 30-meter DEM (Digital Elevation Model) ka slope calculation kaise hota hai?

Answer: Hum SRTM (Shuttle Radar Topography Mission) 30m resolution elevation data ingest karte hain. 3x3 pixel window ke directional gradients dz/dx aur dz/dy se slope angle $\theta = \arctan(\sqrt{(dz/dx)^2 + (dz/dy)^2})$ calculate hota hai. Northeast me >35 degree slopes high-risk category me qualify hoti hain.

Q9. Kya ye system CWC, IMD, GSI aur BRO ke data ke saath integrate ho sakta hai?

Answer: Haan, AlertNex ko modular API integration architecture ke saath design kiya gaya hai. Ye standard GeoJSON, REST Webhooks, aur CAP (Common Alerting Protocol - ITU X.1303) compliant alert payloads export karta hai jo NDMA ke NDEM portal aur state emergency command centres se directly integrate ho sakta hai.

Q10. Low-bandwidth rural internet par frontend ki performance kaisi rehti hai?

Answer: Frontend pure vanilla HTML5, optimized CSS3 tokens aur modular JavaScript me built hai without bulky framework bloat. Assets compressed SVGs me hain, total bundle size < 250KB hai, aur aggressive edge caching ke through 2G/3G connectivity par bhi sub-second load time deliver karta hai.

Q11. Is system ka maintenance aur deployment cost kya hoga state government ke liye?

Answer: AlertNex completely open-source technologies (FastAPI, Leaflet, PostGIS, Linux) par built hai jisse proprietary GIS software licenses ka karodon ka kharcha bachta hai. State-wide cloud hosting serverless architecture par less than ₹20,000 - ₹30,000 per month me operate ho sakti hai.

Q12. Security aur Emergency Dispatch authorization kaise ensure hoti hai?

Answer: System me Role-Based Access Control (RBAC) implement kiya gaya hai with JWT authentication. Public users keval hazard report aur map view kar sakte hain, jabki Siren broadcast, SMS emergency trigger aur route diversion authority keval authenticated District Disaster Commanders ke secure dashboard se hi ho sakti hai.

Q13. Prototype vs Production version me kya farq hai?

Answer: Prototype demonstration version me calibrated synthetic telemetry data aur realistic NER terrain nodes use kiye gaye hain for live hackathon evaluation. Production roll-out me ye direct CWC automated rain gauges, ISRO Bhuvan satellite telemetry feeds aur BRO live road obstruction APIs se connect hoga.

Q14. Northeast India ke tribal communities tak alert kaise pahuchega?

Answer: AlertNex multi-channel alert dispatch support karta hai: (1) Offline SMS Cell-Broadcast gateway (jo smartphones ke bina basic feature phones par kaam karta hai); (2) Local Community Siren Audio Triggers; aur (3) Village Disaster Management Committees (VDMC) WhatsApp / Telegram emergency bots.

Q15. Landslide prediction me soil moisture data itna critical kyu hai?

Answer: Rainfall akele landslide nahi banata — jab rainwater soil pore pressure ko increase karta hai aur cohesion loss hoti hai, tab slope fail hota hai. Soil moisture saturation percentage slope stability limit (factor of safety < 1.0) ko accurate identify karti hai.

Q16. Map view me kaun-kaun se interactive filters available hain?

Answer: GIS map me user Risk Level filters (Low, Moderate, High, Critical), Telemetry Sensor Overlays (Rain gauges, Soil probes, Displacement monitors), Road Network Layer, Village Vulnerability Markers aur Historical Slide hotspots ko dynamically toggle kar sakte hain.

Q17. System failure ya server crash ke time disaster protocol kya hai?

Answer: Hum Multi-Region Edge CDN aur Auto-Failover Cloud Architecture use karte hain. Client-side Service Worker last-known risk state aur evacuation maps ko local cache me preserve rakhta hai even if backend goes down momentarily.

Q18. Kya ye system earthquake-triggered landslides ko bhi factor karta hai?

Answer: Northeast Seismic Zone V me fall karta hai. Hamare historical landslide index aur risk matrix me seismic fault proximity ko baseline hazard coefficient ke roop me integrate kiya gaya hai.

Q19. Simulation Mode ka kya role hai platform me?

Answer: Topbar par 'Demo Simulation' feature disaster commanders aur NDRF teams ko monsoon cloudburst scenarios (e.g. +200mm rainfall in 3 hours) simulate karke contingency evacuation drills run karne ki facility deta hai.

Q20. Government officials ke liye reporting format kya provide hota hai?

Answer: AlertNex one-click automated PDF/Print Incident Dossier generation support karta hai jisme active sector risk, affected road segments, isolated population estimate aur recommended SDRF deployment units ka formal report ready hota hai.

Q21. Frontline workers ke liye mobile compatibility kaise ensure ki gayi hai?

Answer: App me responsive slide-in drawer navigation, touch-optimized minimum 44px tap targets, auto-stacking KPI grids aur adaptive GIS map height integrated hai jo smartphones par seamless work karta hai.

Q22. Future roadmap me Computer Vision drone integration ka kya scope hai?

Answer: Phase 2 roadmap me hum post-slide UAV drone imagery segmentation model integrate karenge jo blocked road length aur debris volume estimation automate karega for faster earth-moving machinery dispatch.

Q23. Team ka background aur technical specialization kya hai?

Answer: Hamari 6-member team me AI/ML specialists, Backend API engineers, GIS cartography developers, UI/UX frontend designers aur Cloud DevOps engineers ka well-balanced mix hai jinhone milkar end-to-end full stack platform build kiya hai.

Q24. MDoNER ke Vision 2047 se AlertNex kaise align hota hai?

Answer: MDoNER ka primary objective resilient infrastructure aur seamless multimodal connectivity provide karna hai. AlertNex mountain lifelines ko climate resilient bana kar trade, tourism aur civilian security ko safeguard karta hai.

Q25. Closing punchline jo judges ko remember rahe?

Answer: *'AlertNex transforms disaster management from reactive rescue into proactive preparedness — because in mountain emergencies, 6 hours of early intelligence saves hundreds of human lives.'*